

Messaging & Notification Sequence Diagrams

MIT ACT — Crennotech Group Project (SiteChat) · Week 4 · 14 August 2026

1. Sending and receiving a message

1. User A types a message in a Site's chat room and taps Send.
2. App writes the message to Firestore under that Site's chat room collection.
3. Firestore triggers a Cloud Function on new message creation.
4. Cloud Function checks who else has that Site's chat room open, and sends a Cloud Messaging push notification to everyone else.
5. User B's app receives the Firestore update in real time and shows the new message; if the app is closed, User B gets a push notification instead.

2. Push notification for a mention

1. User A types a message that includes "@UserB".
2. Cloud Function detects the mention when the message is saved.
3. Cloud Function sends a targeted push notification to User B specifically, marked as a mention (not a general new-message alert).
4. User B taps the notification and is taken directly to that message in the chat room.

3. CleanTasker system message (e.g. "Inspection completed")

1. An inspection is marked complete inside CleanTasker.
2. CleanTasker's backend calls a SiteChat API endpoint (Cloud Function) with the event details.
3. Cloud Function posts a system message into that Site's chat room, tagged differently from user messages (e.g. shown in a distinct style/icon).
4. All users with that Site's chat room open see the update instantly; others receive a push notification if enabled for that event type.